

# Technology Management - NUIST

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## Engineering Technology

May, 2026



## Technology Management - NUIST (A35612)

**Short Title:** Technology Management - NUIST  
**Faculty** Science and Computing  
**Department** Engineering Technology  
**Cluster** Engineering Practise  
**Author** CLAIRE.FITZPATRICK  
**Credits** 5

**Level:** Postgraduate

### Description of Module / Aims

The aim of this module is to prepare individuals with a strategic mind-set, relevant to the biomedical sector of Ireland, fine tuned communication skills, and the ability to drive enhanced performance through the use of technology. Complex case studies involving organisational theory, conflict resolution and effective strategic management are core to this module.

### Programmes

|           |                                                                     |           |
|-----------|---------------------------------------------------------------------|-----------|
| ENGR-0040 | Certificate in Biomedical Engineering (WD_CBIOM_MA)                 | 1 / 2 / M |
| ENGR-0040 | MEng in Electronic Engineering (WD_EELEN_R)                         | 1 / 2 / E |
|           | MEng in Electronic Information Systems (WD_EELEN_RI)                | 3 / 6 / M |
| ENGR-0040 | MSc in Innovative Technology Engineering (WD_CINNT_R)               | 1 / 2 / M |
| ENGR-0040 | PG Dip in Science in Innovative Technology Engineering (WD_CINNT_G) | 1 / 2 / M |

### Indicative Content

- Engineering and Technology Management: The management process: Empowerment and effective leadership
- Engineering and Technology Management: The regional & Global Context: Sources of competitive advantage: Insights into professional imperatives
- Ethics in Engineering / Technology
- Intellectual Property: Creating, Application and Protection: Learning Organisations
- Manufacturing and Production Management: Strategies and Tools
- Manufacturing & Production Management: Strategic importance of maintenance and reliability
- Career Development in a technology company. Managing your boss, peers and subordinates

### Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Using Kotter's Eight Step Model, synthesise change management tools specific to the optimisation of R&D processes in high value organisations.
2. Apply the knowledge and comprehension gained in the analysis of complex Enterprise Technology Management case studies to provide best in-class frameworks for SME and MNC organisations.
3. Evaluate complete and/or partial management strategies in respect of the research, design and manufacture of modern technological products; both with and without the aid of industry standard software support tools.
4. Synthesise complete and/or partial management strategies in respect of the research, design and manufacture of modern technological products; both with and without the aid of industry standard software support tools.

## Learning and Teaching Methods

- Lectures
- Case Studies
- Assignments / Assessments
- Problem Based Mini Project

## Assessment Methods

|                       | Weighing | Outcomes Assessed |
|-----------------------|----------|-------------------|
| Continuous Assessment | 100%     |                   |
| Group Project         | 40%      | 1,2,3,4           |
| Essay                 | 50%      | 1,2,3,4           |
| Participation         | 10%      | 1,2               |

## Assessment Criteria

70% -

**100%:** The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

**60% - 69%:** The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

**40% - 59%:** The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

**<40%:** The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 24        |           |
| Practical            |           |           |
| Independent Learning | 111       |           |

## Essential Material

- Griffin, R.W. *Management*. USA: Houghton Mifflin, 2008.

## Supplementary Material

- Betz, F. *Strategic Technology Management*. New York: McGraw Hill, 1997.
- Tavani, T.H. *Ethics & Technology*. 2nd ed.. USA: Wiley, 2007.